A SHORT CLASSIFICATION OF COMPLEX RANK-TWO GROTHENDIECK RINGS

Abstract. We give a short proof that a complex rank-two fusion category has Grothendieck ring $X^2 = 1 + nX$ with $n = 0$ or 1. A norm argument in its sphericalization first establishes sphericality without assuming pivotality. The center's Gauss sum then excludes $n \geq 2$ by a two-term triangle inequality. The argument uses standard structural results for fusion categories and does not classify tensor structures.

1. Introduction

Ostrik classified the complex fusion categories of rank two, including their tensor structures [4]. We give a compact proof at the level of Grothendieck rings. The new proof separates the existence of a spherical structure from the fusion-rule exclusion and does not assume a braiding on the original category. For a fusion category with simples $\mathbf{1}, X$, rigidity gives $X^* = X$ and

$$X^2 = \mathbf{1} + nX, \qquad n \in \mathbb{Z}_{\geq 0}. \tag{1}$$

Write K_n for this based ring.

Theorem 1.1. *For a fusion category $\mathcal{C}$ over $\mathbb{C}$ with exactly two simple objects, $K_0(\mathcal{C})$ is isomorphic to K_0 or K_1. Every such category admits a spherical structure. Both rings occur, respectively in a pointed category and in a Fibonacci category.*

The statement concerns fusion rings, rather than the monoidal-equivalence classification of their realizations. In particular, the two categories with ring K_0 and the two with ring K_1 in [4] need not be identified to prove Theorem 1.1.

2. A spherical structure from a norm

We recall the general results used below. A fusion category over $\mathbb{C}$ admits a *sphericalization* $\pi : \mathcal{S} \to \mathcal{C}$, where $\mathcal{S}$ is spherical. Every simple V has two simple lifts V_+, V_-; the lifts of $\mathbf{1}$ are $\mathbf{1}$ and a central invertible object ε of dimension -1, and $\varepsilon \otimes V_+ = V_-$ [1, Remark 3.1]. If V is self-dual, so are both lifts [5, Section 5.1]. We use the general *formal-codegree divisibility theorem*: if f is a formal codegree of a complex fusion category $\mathcal{A}$, then $\dim(\mathcal{A})/f$ is an algebraic integer [5, Corollary 2.14]. The global dimension here is the intrinsic global dimension and does not require a pivotal structure.

Lemma 2.1. *Suppose $n \geq 1$ in (1). If x is either lift of X in the sphericalization, its signed fusion rule modulo $\varepsilon = -1$ is*

$$x^2 = 1 + \widetilde{n}x, \qquad \widetilde{n} \in \mathbb{Z}, \quad |\widetilde{n}| \leq n, \quad \widetilde{n} \equiv n \pmod 2. \tag{2}$$

Moreover $|\widetilde{n}| = n$.

Proof. As x is self-dual, its square contains $\mathbf{1}$ exactly once and ε zero times. The other constituents have the form $rx + s\varepsilon x$ with $r, s \geq 0$ and $r + s = n$. Reducing modulo $\varepsilon = -1$ gives $\widetilde{n} = r - s$, proving the first assertions.

Let $d = \dim(x)$. The spherical dimension character kills $\varepsilon + 1$, so $d^2 = 1 + \widetilde{n}d$. The global dimension of $\mathcal{C}$ equals that of one choice of lift squared, namely

$$\delta = 1 + d^2; \tag{3}$$

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the global dimension of $\mathcal{S}$ is 2δ. Put $R = \mathbb{Q}[t]/(t^2 - nt - 1)$. Its two formal codegrees are $f_\pm = 1 + u_\pm^2$, where $u_\pm = (n \pm \sqrt{n^2+4})/2$. Consequently

$$N_{\mathbb{Q}(\sqrt{n^2+4})/\mathbb{Q}}(f_+) = n^2 + 4. \tag{4}$$

The number $n^2 + 4$ is nonsquare for $n \geq 1$, so R is a quadratic field. The two lifts of each simple double each codegree of R in $K_0(\mathcal{S})$. Thus $2f_+$ is a formal codegree of $\mathcal{S}$, and general codegree divisibility says that δ/f_+ is an algebraic integer.

To take its norm in a single quadratic field, we recall a second general formal-codegree fact: the codegrees of a spherical category belong to the field generated by its simple dimensions [5, Corollary 2.15]. This field is $L = \mathbb{Q}(d)$, of degree at most two, and it contains both roots $2f_\pm$ (the conjugate codegrees of $\mathcal{S}$). Hence $L = \mathbb{Q}(\sqrt{n^2+4})$; it is also the field of $\widetilde{n}$'s quadratic character, so $\widetilde{n} \neq 0$. The conjugate of δ is $1 + d'^2$ with d' the other root of $t^2 - \widetilde{n}t - 1$. Taking norms in L yields

$$N_{L/\mathbb{Q}}(\delta/f_+) = \frac{\widetilde{n}^2 + 4}{n^2 + 4} \in \mathbb{Z}_{>0}.$$

The ratio is positive and at most one by $|\widetilde{n}| \leq n$. Therefore it equals one and $|\widetilde{n}| = n$. □

Corollary 2.2. *Every complex rank-two fusion category admits a spherical structure.*

Proof. For $n = 0$ the category is pointed and hence pseudo-unitary; its canonical pivotal structure is spherical [1, Propositions 8.23–8.24]. For $n \geq 1$, changing x to εx changes $\widetilde{n}$ to $-\widetilde{n}$. By Lemma 2.1 we may choose $\widetilde{n} = n$. Since $r + s = n$ and $r - s = n$, we have $s = 0$: $x^2 = \mathbf{1} \oplus nx$ already inside $\mathcal{S}$. The full replete subcategory with simples $\mathbf{1}, x$ is closed under tensor products and duals. The forgetful tensor functor π restricts to a tensor equivalence from it to $\mathcal{C}$. Restricting and transporting the spherical structure of $\mathcal{S}$ proves the claim. □

3. The center's two known summands

Let $\mathcal{C}$ now have the spherical structure of Corollary 2.2, write $d = \dim(X)$, and set $D = \dim(\mathcal{C}) = 1 + d^2$. We recall two standard properties of its modular Drinfeld center $\mathcal{Z}(\mathcal{C})$:

$$\sum_{B \in \operatorname{Irr} \mathcal{Z}(\mathcal{C})} \dim(B)^2 = D^2, \qquad \sum_{B \in \operatorname{Irr} \mathcal{Z}(\mathcal{C})} \theta_B \dim(B)^2 = D. \tag{5}$$

The first follows from $\dim \mathcal{Z}(\mathcal{C}) = D^2$ and the second is the center Gauss-sum identity [3, Theorem 1.2]. For completeness, the second identity follows by writing $\dim(B) = \sum_V [F(B) : V] \dim(V)$, using adjunction, and summing $\dim(V) \operatorname{Tr}(\theta_{I(V)})$ over the simple objects V: the twist trace is D for $V = \mathbf{1}$ and zero otherwise [5, Theorem 2.5]. Here θ_B is the canonical twist; its values are roots of unity by the Vafa theorem.

Let $F : \mathcal{Z}(\mathcal{C}) \to \mathcal{C}$ be forgetful and I its right adjoint. The standard formula $FI(\mathbf{1}) = \bigoplus_V V \otimes V^*$ gives $FI(\mathbf{1}) = 2\mathbf{1} \oplus nX$. By adjunction, $\dim_{\mathbb{C}} \operatorname{End}_{\mathcal{Z}(\mathcal{C})} I(\mathbf{1}) = 2$. The simple unit occurs once in $I(\mathbf{1})$, so semisimplicity implies

$$I(\mathbf{1}) = \mathbf{1} \oplus A, \qquad F(A) = \mathbf{1} \oplus nX, \qquad \dim(A) = 1 + nd = d^2. \tag{6}$$

The twist of each summand of $I(\mathbf{1})$ is one by the same trivial-twist theorem [5, Theorem 2.5]; equivalently, the canonical connected commutative separable algebra $I(\mathbf{1})$ has trivial twist.

Proof of Theorem 1.1. The sphericality assertion is Corollary 2.2. Assume $n \geq 1$. The two roots of $d^2 = 1 + nd$ are $d_\pm = (n \pm \sqrt{n^2+4})/2$. Choose an automorphism σ of $\overline{\mathbb{Q}}$ sending the actual categorical dimension d to $d_+ > 0$. For any simple central object B, its dimension is the dimension of $F(B) = a_B\mathbf{1} \oplus b_B X$ with $a_B, b_B \in \mathbb{Z}_{\geq 0}$, and hence $\sigma(\dim(B)) = a_B + b_B d_+ > 0$. Applying σ to (5) preserves the equalities and sends each twist to another root of unity; no Galois descent of associators is involved.

Set $t = d_+^2$ and $D_+ = 1 + t$. The two objects in (6) contribute $1 + t^2$ to both identities in (5). Thus the other simple central objects satisfy

$$\sum_{B \neq \mathbf{1}, A} \sigma(\dim(B))^2 = D_+^2 - 1 - t^2 = 2t, \qquad \sum_{B \neq \mathbf{1}, A} \sigma(\theta_B)\sigma(\dim(B))^2 = D_+ - 1 - t^2 = t(1-t).$$

All coefficients in the first sum are nonnegative real numbers and $|\sigma(\theta_B)| = 1$. The triangle inequality gives $t|1 - t| \leq 2t$, so $t \leq 3$. For $n \geq 2$, $d_+ > n \geq 2$ and therefore $t > 4$, a contradiction. Hence $n \leq 1$. Finally K_0 is realized by $\mathrm{Vec}_{\mathbb{Z}/2}$ and K_1 by the Fibonacci fusion category, for instance the even part of the $SU(2)_3$ fusion category [4]. Both are spherical. □

Remark 3.1 (Positive characteristic). The argument above is specific to characteristic zero. Etingof, Ostrik and Venkatesh [2, Propositions 4.6–4.7] prove that over an algebraically closed field of positive characteristic no *braided* fusion category realizes K_n for $n \geq 2$, and explicitly leave the general fusion-category case open. In particular their result cannot be used to assert a full positive-characteristic classification. A category of zero global dimension in characteristic p need not have a semisimple modular center, so the proof of (5) does not carry over directly. We know of no later general resolution of the nonbraided rank-two problem in positive characteristic.

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